

Homework 3

1. Problem 2.1

Why does the following identity hold?

$$\nabla_{\mathbf{w}}(\mathbf{w}^T \mathbf{A} \mathbf{w}) = (\mathbf{A} + \mathbf{A}^T) \mathbf{w}$$

For the 2D case, let

$$\mathbf{A} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, \mathbf{w} = \begin{bmatrix} w_1 \\ w_2 \end{bmatrix}$$

Expanding the quadratic form gives

$$\begin{aligned} \mathbf{w}^T \mathbf{A} \mathbf{w} &= \begin{bmatrix} w_1 & w_2 \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} \\ &= \begin{bmatrix} w_1 a + w_2 c & w_1 b + w_2 d \end{bmatrix} \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} \\ &= (w_1 a + w_2 c) w_1 + (w_1 b + w_2 d) w_2 \\ &= w_1^2 a + w_1 w_2 c + w_1 w_2 b + w_2^2 d \end{aligned}$$

Therefore,

$$\begin{aligned} \nabla_{\mathbf{w}}(\mathbf{w}^T \mathbf{A} \mathbf{w}) &= \begin{bmatrix} \frac{\partial}{\partial w_1} w_1^2 a + w_1 w_2 c + w_1 w_2 b + w_2^2 d \\ \frac{\partial}{\partial w_2} w_1^2 a + w_1 w_2 c + w_1 w_2 b + w_2^2 d \end{bmatrix} \\ &= \begin{bmatrix} 2w_1 a + w_2 c + w_2 b \\ w_1 c + w_1 b + 2w_2 d \end{bmatrix} \\ &= \begin{bmatrix} 2w_1 a + w_2(b + c) \\ w_1(b + c) + 2w_2 d \end{bmatrix} \end{aligned}$$

Notice that

$$\begin{aligned} \mathbf{A} + \mathbf{A}^T &= \begin{bmatrix} a & b \\ c & d \end{bmatrix} + \begin{bmatrix} a & c \\ b & d \end{bmatrix} \\ &= \begin{bmatrix} 2a & b + c \\ b + c & 2d \end{bmatrix} \end{aligned}$$

So,

$$\begin{aligned} (\mathbf{A} + \mathbf{A}^T) \mathbf{w} &= \begin{bmatrix} 2a & b + c \\ b + c & 2d \end{bmatrix} \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} \\ &= \begin{bmatrix} 2w_1 a + w_2(b + c) \\ w_1(b + c) + 2w_2 d \end{bmatrix} \end{aligned}$$

Therefore,

$$\nabla_{\mathbf{w}}(\mathbf{w}^T \mathbf{A} \mathbf{w}) = (\mathbf{A} + \mathbf{A}^T) \mathbf{w}$$

Showing that the identity holds.